

NeuroSense Three-Device Setup Guide

Plain-English setup, testing flow, and video demo checklist for the Tab5 wall unit, ESP32 environment hub, and BHA2 RR/HR bed sensor.

BHA2 mmWave RR/BR, HR, presence	Tab5 Wall Unit Mood emoji check-in	ESP32 Env Hub Temperature, humidity, air quality	NeuroSense Dashboard Combines everything by bed
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Simple meaning: three small devices send different information to one dashboard. The dashboard knows the data belongs together because all three devices use the same facility and bed.

1. What we are building

In plain terms, NeuroSense is a care-room monitoring system. One bed has three devices around it. Each device has one clear job, and the dashboard joins their data into one bed view.

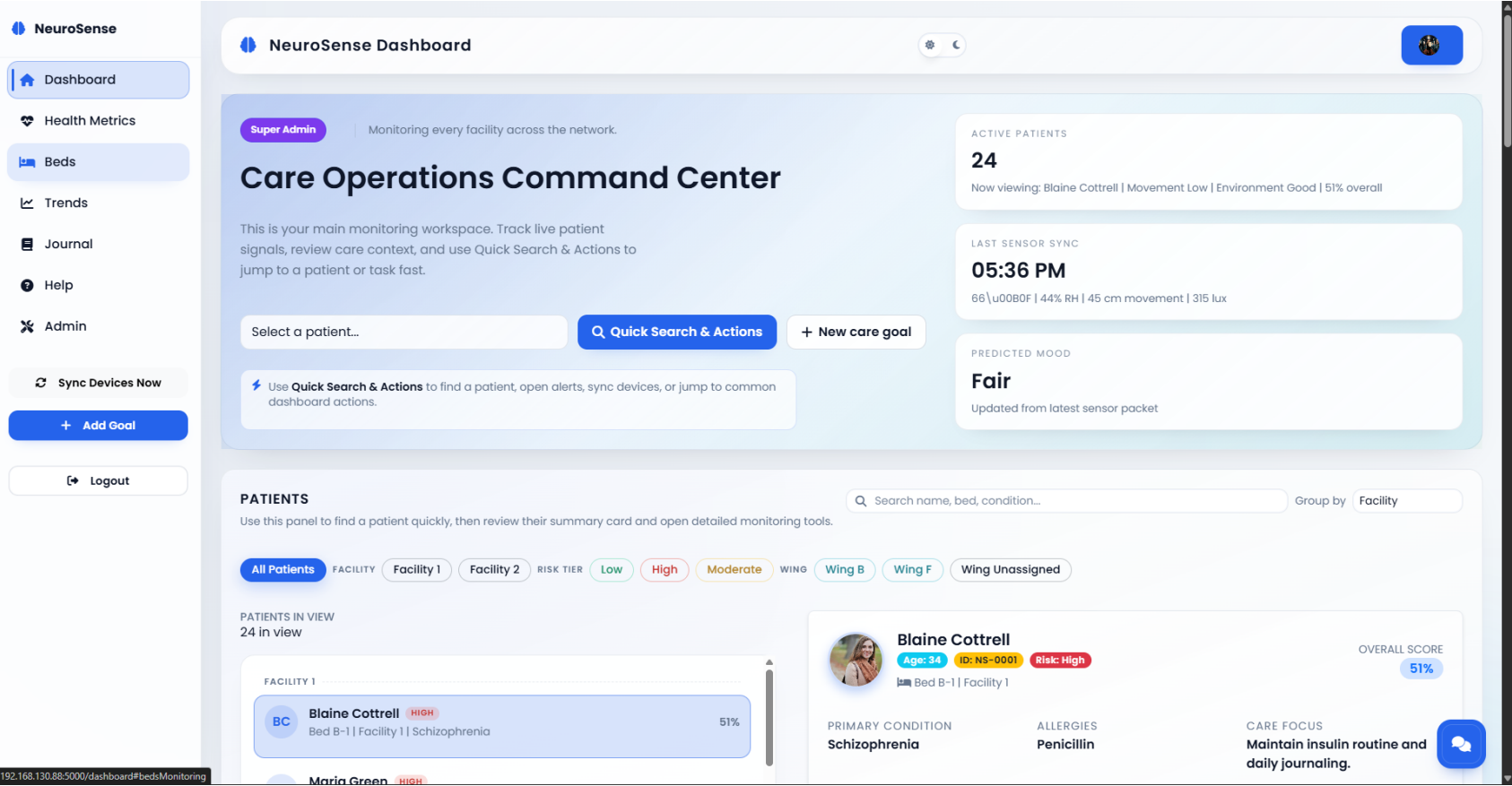
- **BHA2 / mmWave bed sensor:** watches the patient from the correct position and sends breathing rate, heart rate, presence, confidence, and distance.
- **M5Stack Tab5 wall unit:** shows the mood emoji screen. The patient taps how they feel, and the wall unit sends that mood to the dashboard.
- **ESP32 environment hub:** reads room conditions such as temperature, humidity, air quality, light, noise, and motion.
- **Dashboard:** receives all device messages, stores them, displays trends, and raises alerts when needed.

The devices do not need to talk directly to each other. They all talk to the backend. The backend joins their data using the same facility_id and bed_id.

Device-to-dashboard map

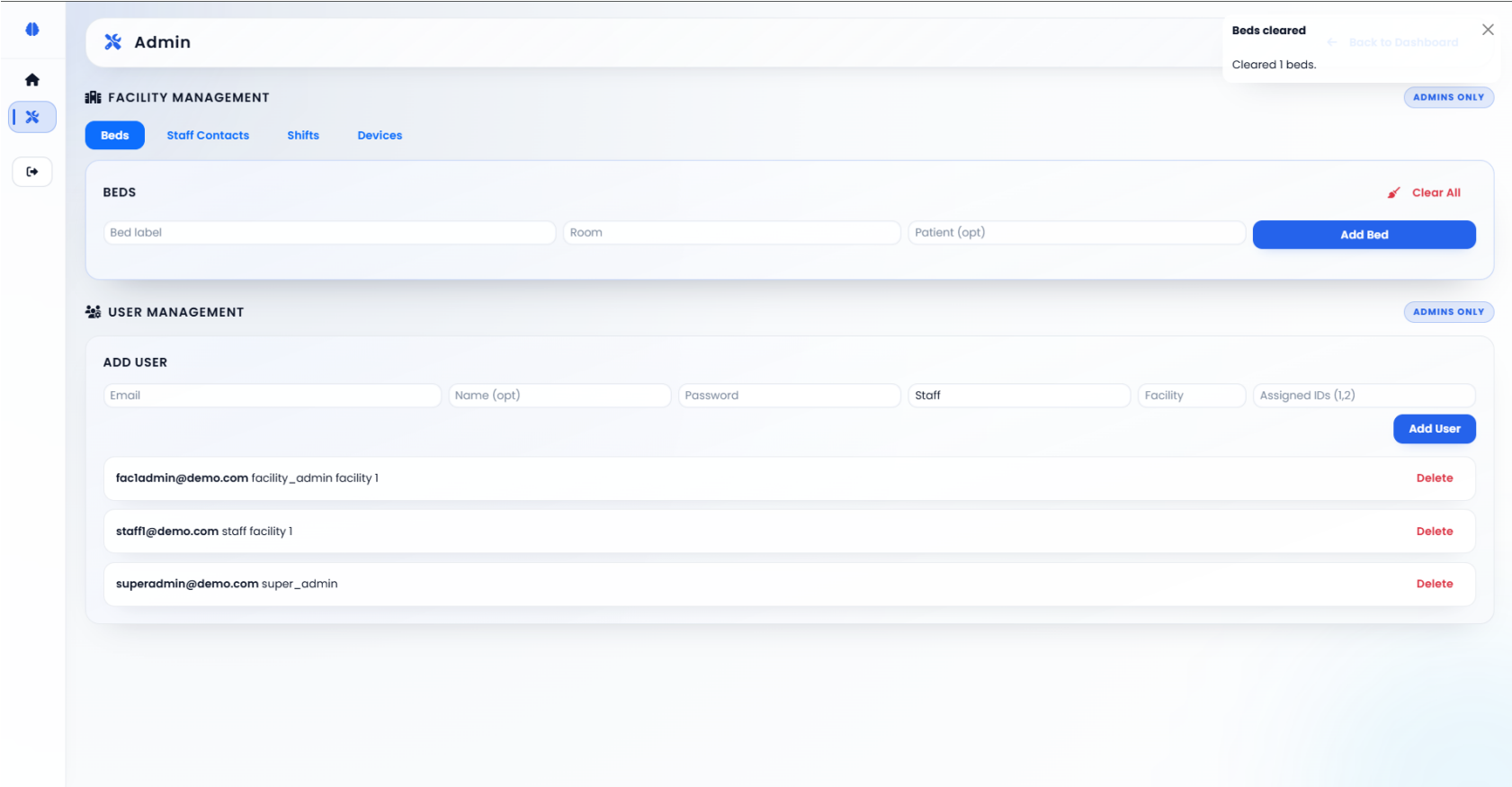
Device	Example ID	Endpoint	What it sends
BHA2 bed sensor	BHA2-001	/api/v1/telemetry	RR/BR, HR, presence, distance, confidence
Tab5 wall unit	WALL-001	/api/v1/wall/event	mood_checkin
ESP32 environment hub	ENV-001	/api/v1/wall/event	temperature, humidity, air quality, light, noise/motion

Screenshot 1



Dashboard home: this is the provider command center where the final combined data will be shown.

Screenshot 2



Admin > Beds: create a bed first. The bed label is the human-friendly name; the long Bed ID is the technical ID devices should use.

Screenshot 3

Admin

FACILITY MANAGEMENT

Beds | Staff Contacts | Shifts | Devices

BEDS

Clear All

Bed label Room Patient (opt) Add Bed

BED-01 | a4045022-03c4-4675-8eb1-6e2ede4e19ac

USER MANAGEMENT

ADD USER

Email Name (opt) Password Staff Facility Assigned IDs (1,2) Add User

fac1admin@demo.com	facility_admin	facility 1	Delete
staff1@demo.com	staff	facility 1	Delete
superadmin@demo.com	super_admin		Delete

Admin > Beds after creation: BED-01 now exists and has a long bed UUID. Copy the correct bed ID when configuring firmware.

Screenshot 4

The screenshot shows the NeuroSense Admin interface. The top navigation bar includes a 'Back to Dashboard' link. The main section is titled 'Admin' and contains a 'FACILITY MANAGEMENT' section with tabs for 'Beds', 'Staff Contacts', 'Shifts', and 'Devices'. The 'Devices' tab is active, showing a list of devices. Each device entry includes a name, a bed ID, a device type, and a status. The 'BHA2-001' device is highlighted, showing its details and a 'Latest Device API Key' section with a redacted key. The 'ADD USER' section is visible at the bottom.

Admin [Back to Dashboard](#)

FACILITY MANAGEMENT [ADMINS ONLY](#)

[Beds](#) [Staff Contacts](#) [Shifts](#) [Devices](#)

DEVICES [Clear All](#)

DEV-001 Bed ID Telemetry sensor Firmware [Add Device](#)

Latest Device API Key
Device created for BHA2-001. Copy this now. It cannot be viewed again later. [Copy Key](#) [Dismiss](#)

DEVICE API KEY REDACTED - copy real key from your dashboard only once

BHA2-001 [BHA2 SENSOR](#) [ASSIGNED](#)
Bed a4045022-03c4-4675-8eb1-6e2ede4e19ac · FW --
[presence](#) [rr](#) [hr](#)

ENV-001 [WALL UNIT](#) [ASSIGNED](#)
Bed a4045022-03c4-4675-8eb1-6e2ede4e19ac · FW --
[mood](#) [environment](#)

WALL-001 [WALL UNIT](#) [ASSIGNED](#)
Bed a4045022-03c4-4675-8eb1-6e2ede4e19ac · FW tab5_wall_unit_v6
[mood](#) [environment](#)

USER MANAGEMENT [ADMINS ONLY](#)

ADD USER

Email	Name (last)	Username	Staff	Facility	Assigned (De (1,2))
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Admin > Devices: BHA2-001, ENV-001, and WALL-001 are all assigned to the same bed. The device API key is redacted in this guide.

Security note: never share the real device API key in screenshots, PDFs, or videos. It is shown once in the admin UI so you can copy it into the correct firmware secret file.

2. Setup steps - from empty dashboard to working bed

Follow this order. Do not start with the devices. First create the bed and device records in the dashboard, then copy the IDs into the firmware.

Step 1 - Start the backend

Run the Flask dashboard on your laptop/server. For physical devices to reach it, the app must listen on the network, not only on localhost.

Step 2 - Create the bed

Go to Admin > Beds. Add a bed label such as BED-01. After creation, copy the long Bed ID shown beside the label.

Step 3 - Add three devices

Go to Admin > Devices. Create BHA2-001, ENV-001, and WALL-001. Assign each one to the same Bed ID.

Step 4 - Copy each API key once

Each device gets its own bearer key. Copy it immediately into that device firmware secrets file.

Step 5 - Upload firmware one at a time

Upload ESP32 environment hub first in dummy mode, then Tab5 wall unit, then BHA2 telemetry.

Step 6 - Check the dashboard

Open the Beds section and confirm mood, environment, and RR/HR are appearing under the same bed.

Important: Bed label and Bed ID are not always the same thing

In the screenshots, the bed label is BED-01, but the dashboard also shows a long technical Bed ID like a4045022-03c4-4675-8eb1-6e2ede4e19ac. For firmware, use whatever your backend expects for `bed_id`. If your devices are rejecting with `bed/facility` mismatch, this is the first thing to check.

3. Backend and network setup

When testing on your laptop, the ESP32, Tab5, and BHA2 cannot use localhost. Localhost means "myself". To a device, localhost is the device itself, not your laptop.

Run Flask so it is reachable from other devices:

```
flask run --host=0.0.0.0 --port=5000
```

Then use your laptop/server IP address in firmware:

```
# Example only - use your actual IP
#define SERVER_BASE_URL "http://192.168.1.50:5000"
```

Before recording, confirm these are true:

- All devices and laptop/server are on the same Wi-Fi/network.
- The Wi-Fi is 2.4 GHz for ESP32 devices.
- Windows Firewall allows inbound traffic to port 5000.
- Each firmware uses the correct SERVER_BASE_URL, device_id, facility_id, bed_id, and API key.
- The backend terminal is open so you can see incoming requests.

4. What to put in each firmware secrets file

Each device has its own secret file. Think of it as that device's ID card. The device uses it to connect to Wi-Fi, find the dashboard, and prove it is allowed to send data.

ESP32 environment hub secrets.h

```
#define WIFI_SSID      "YOUR_2_4GHZ_WIFI_NAME"
#define WIFI_PASSWORD  "YOUR_WIFI_PASSWORD"
#define SERVER_BASE_URL "http://YOUR_SERVER_IP:5000"
#define ENV_API_KEY    "PASTE_ENV_DEVICE_API_KEY_HERE"
#define ENV_DEVICE_ID  "ENV-001"
#define FACILITY_ID    "FAC123"
#define BED_ID         "PASTE_THE_REAL_BED_ID_HERE"
```

Tab5 wall unit secrets.h

```
#define WIFI_SSID      "YOUR_2_4GHZ_WIFI_NAME"
#define WIFI_PASSWORD  "YOUR_WIFI_PASSWORD"
#define SERVER_BASE_URL "http://YOUR_SERVER_IP:5000"
#define WALL_API_KEY   "PASTE_WALL_DEVICE_API_KEY_HERE"
#define WALL_DEVICE_ID "WALL-001"
#define FACILITY_ID    "FAC123"
#define BED_ID         "PASTE_THE_REAL_BED_ID_HERE"
```

BHA2 bed sensor secrets.h

```
#define WIFI_SSID      "YOUR_2_4GHZ_WIFI_NAME"
#define WIFI_PASSWORD  "YOUR_WIFI_PASSWORD"
#define API_URL        "http://YOUR_SERVER_IP:5000/api/v1/telemetry"
#define DEVICE_API_KEY "PASTE_BHA2_DEVICE_API_KEY_HERE"
#define DEVICE_ID      "BHA2-001"
#define FACILITY_ID    "FAC123"
#define BED_ID         "PASTE_THE_REAL_BED_ID_HERE"
```

Use one API key per device. Do not reuse the BHA2 key for the Tab5 or ESP32 environment hub unless the backend is deliberately configured that way.

5. Testing each device before the full video

Test the devices one by one first. This makes debugging much easier.

Test environment endpoint with curl

```
curl -i -X POST "http://localhost:5000/api/v1/wall/event" \
  -H "Authorization: Bearer PASTE_ENV_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{
    "device_id": "ENV-001",
    "facility_id": "FAC123",
    "bed_id": "PASTE_THE_REAL_BED_ID_HERE",
    "event_type": "environment",
    "source": "esp32_environment_hub",
    "temperature_c": 24.3,
    "humidity": 67,
    "air_quality": "OK",
    "light_level": 120,
    "noise_level": 34
  }'
```

Expected result: HTTP 200/201 and a JSON response with **"ok": true**.

Test BHA2 endpoint with curl

```
curl -i -X POST "http://localhost:5000/api/v1/telemetry" \
  -H "Authorization: Bearer PASTE_BHA2_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{
    "device_id": "BHA2-001",
    "facility_id": "FAC123",
    "bed_id": "PASTE_THE_REAL_BED_ID_HERE",
    "presence": true,
    "rr": 17,
    "hr": 84,
    "confidence": 0.9,
    "firmware": "bha2_demo",
    "capabilities": ["presence", "rr", "hr", "distance"]
  }'
```

Expected result: telemetry is stored and the dashboard bed view starts showing RR/HR trends.

6. Full three-device demo flow

This is the order to use when you want to record the system working end-to-end.

1. Start Flask backend and keep the terminal visible.
2. Open the dashboard in the browser and go to Admin > Devices to show WALL-001, ENV-001, and BHA2-001 assigned to the same bed.
3. Power the ESP32 environment hub. In dummy mode it should post environment values every 30 seconds.
4. Tap a mood emoji on the Tab5. The Tab5 should show Sending, then Saved.
5. Run the BHA2 device. Serial Monitor should show breath_rate, heart_rate, distance, and HTTP success.
6. Return to the dashboard bed view and show mood, environment values, and RR/HR together.

What success looks like

Part	Success sign
ESP32 environment hub	Serial Monitor shows HTTP 200/201 for /api/v1/wall/event. Dashboard shows temperature/humidity/air quality.
Tab5 wall unit	Screen shows Saved after tapping a mood face. Dashboard shows latest mood.
BHA2 sensor	Serial Monitor shows RR/HR and HTTP 200/201. Dashboard shows RR/HR trend.

7. Video recording script

Use this short script so the video sounds organized and easy to follow.

Intro:

"This is the NeuroSense room monitoring setup. We have three devices connected to one bed: the BHA2 for breathing and heart rate, the Tab5 wall unit for patient mood check-ins, and an ESP32 environment hub for room conditions."

Admin setup:

"All three devices are assigned to the same facility and bed. That is how the backend knows the data belongs together."

Environment hub:

"The ESP32 is currently in dummy mode. It is sending test temperature, humidity, air quality, light, and noise values so we can prove the dashboard pipeline before wiring sensors."

Tab5 mood:

"The patient taps a face on the wall unit. The screen sends the mood to the backend and shows Saved when the dashboard accepts it."

BHA2 vitals:

"The BHA2 sensor sends respiration rate and heart rate into the telemetry endpoint. These values appear in the bed monitoring dashboard."

Closing:

"Now one bed has body data, room data, and patient-reported mood coming into one dashboard view."

What to avoid showing in the video

- Do not show full API keys. Blur or cover them.
- Do not show real patient information in a public demo.
- Use demo bed IDs and demo patient names when recording for investors or public updates.

8. Common problems and quick fixes

Problem	Fix
Device says offline or cannot post	Use SERVER_BASE_URL with your laptop/server IP, not localhost. Check that Flask is running on 0.0.0.0 and Windows Firewall allows port 5000.
HTTP 401 or 403	The API key is wrong, missing, copied with spaces, or belongs to another device. Generate/copy the key again from Admin > Devices.
HTTP 400 bed/facility mismatch	The device_id, facility_id, or bed_id in firmware does not match the dashboard device assignment. Use the real bed ID shown in Admin > Beds.
Tab5 Wi-Fi code does not compile	Keep the UI version that works, then fix the Arduino/M5Stack Tab5 environment separately. The Tab5 uses its own M5Stack board package and Wi-Fi setup.
ESP32 posts but dashboard does not update	Check /api/bed_bundle for the bed. If the data is there, the backend is storing it and the dashboard UI refresh may need attention.
BHA2 gives RR but no HR sometimes	Heartbeat is harder to detect than breathing. Keep the sensor aimed properly, within range, and ask the subject to stay still for 30-60 seconds.

Final reminder: prove the pipeline with dummy data first. Once the dashboard is receiving and displaying data correctly, replace dummy values with real sensor readings one sensor at a time.